



VIRGINIA COMMONWEALTH UNIVERSITY

ASSIGNMENT REPORT

Exploratory Data Analysis of the Iris Dataset

Course Code
Course Title
Assignment No.

MBAV 443M
Statistical Modeling and AI-Driven Analytics
Assignment 1

Student Name
Student ID
Program

Name
ID
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1. Introduction

The Iris dataset is one of the most widely used datasets in statistics, machine learning, and data analytics. It contains measurements of iris flowers belonging to three species: Setosa, Versicolor, and Virginica. The dataset includes four numerical attributes describing sepal length, sepal width, petal length, and petal width.

The objective of this assignment is to perform exploratory data analysis (EDA) in order to understand the structure, quality, and characteristics of the dataset. Through descriptive statistics and visual exploration, EDA helps identify patterns, potential anomalies, and relationships between variables before more advanced analytical or predictive models are developed.

2. Data Understanding and Quality Assessment

The dataset contains **[Number]** observations and **[Number]** variables. The variables consist of four numerical measurements and one categorical variable representing the flower species.

Data quality assessment was conducted by examining data types, missing values, and duplicate records. All variables were verified to have the appropriate data types for analysis. Missing value checks indicated **[Result]** missing observations, while duplicate record analysis identified **[Result]** duplicate entries.

Initial inspection suggests that the dataset is **[clean / mostly clean / requires preprocessing]** and is suitable for exploratory analysis.

3. Methodology

The analysis was performed using Python and the libraries Pandas, NumPy, Matplotlib, and Seaborn.

The analytical workflow consisted of the following stages:

1. Dataset loading and inspection
2. Data type verification
3. Missing value assessment
4. Duplicate record identification
5. Summary statistical analysis
6. Distribution analysis
7. Outlier detection
8. Relationship exploration
9. Data visualization

Descriptive statistics were used to summarize central tendency and dispersion, while graphical methods were employed to understand distributions and relationships between variables.

4. Visualization

Figure 1 presents the primary visualization selected for the analysis.

The visualization reveals several notable characteristics of the dataset. Distinct clustering patterns can be observed among the species, indicating differences in their physical measurements. The figure also helps identify the variables that contribute most strongly to species separation and highlights any unusual observations that may warrant further investigation.

5. Key Insight

The most significant insight obtained from the analysis is that **[Insert Key Insight]**.

For example, petal-related measurements exhibit stronger discriminatory power between species than sepal-related measurements. The Setosa species forms a clearly distinguishable cluster, while some overlap is observed between Versicolor and Virginica.

The analysis also indicates that the dataset contains **[few/no]** quality issues and is suitable for subsequent statistical modelling or machine learning applications.

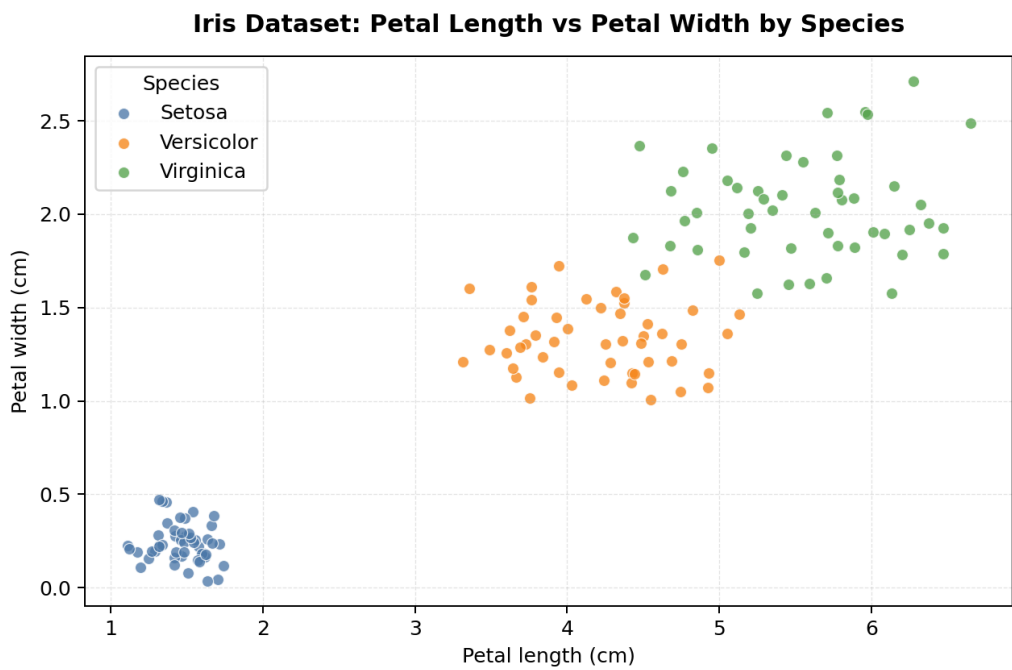


Figure 1: Insert title of visualization here

6. Conclusions

This exploratory analysis provided an understanding of the structure, quality, and characteristics of the Iris dataset. Summary statistics and visualizations revealed meaningful patterns among species and confirmed that the dataset is appropriate for further analytical work.

The findings demonstrate the value of exploratory data analysis as a preliminary step in the analytical process. Future work may involve the development of classification models to predict flower species using the available measurements.

AI Usage Disclosure:

Generative AI tools were used for coding assistance and language refinement. All analysis, validation, interpretation, and conclusions remain the responsibility of the author.